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10⁰⁰-12⁰⁰
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Seriegrafia 3

SERII DE NUMERE REALE PART. 2

(21) $\sum_{n=1}^{\infty} n e^{-(n^2+n)}$

Vom aplica criteriul raportului (*)

$$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to+\infty} \frac{(n+1)\cdot e^{-(n+1)(n+2)}}{n\cdot e^{-n(n+1)}} = \lim_{n\to+\infty} \underbrace{\frac{n+1}{n}}_{\to 1} \cdot e^{n(n+1)-(n+1)(n+2)}$$

$$= \lim_{n\to+\infty} e^{(n+1)(n-n-2)} = \lim_{n\to+\infty} e^{\underbrace{-2(n+1)}_{\to -\infty}} = 0$$

$\underbrace{\quad}_{\to 0}$ (*)

$\Rightarrow$ Cum $\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = 0 < 1 \overset{(*)}{\Rightarrow}$ seria este convergentă.

(22) $\sum_{n=1}^{\infty} \frac{(an)^n}{n!}$

Vom aplica criteriul raportului. (*)

$u_n = \frac{(an)^n}{n!}$

$$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to+\infty} \frac{[a(n+1)]^{n+1}}{(n+1)!} \cdot \frac{n!}{(an)^n} = \lim_{n\to+\infty} \left[\frac{a(n+1)}{an}\right]^n \cdot \frac{n!}{n!(n+1)} \cdot a^{(n+1)}$$

$$= \lim_{n\to+\infty} \left(1+\frac{1}{n}\right)^n \cdot a = a\cdot e$$

Dacă $a\cdot e < 1 \Leftrightarrow a < \frac{1}{e} \overset{(*)}{\Rightarrow}$ seria este convergentă.

Dacă $a\cdot e > 1 \Leftrightarrow a > \frac{1}{e} \overset{(*)}{\Rightarrow}$ seria este divergentă.

Dacă $a\cdot e = 1 \Leftrightarrow a = \frac{1}{e}$

Metoda 1: Vom folosi formula lui Stirling: $n! \simeq \left(\frac{n}{e}\right)^n \cdot \sqrt{2\pi n}$

$$\Rightarrow u_n = \left(\frac{1}{e}\right)^n \cdot \frac{n^n}{n!} = \frac{1}{e^n} \cdot n^n \cdot \frac{e^n}{n^n} \cdot \frac{1}{\sqrt{2\pi n}} = \frac{1}{\sqrt{2\pi n}}, \text{ dar}$$

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{2\pi n}} \sim \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$, care este serie divergentă (serie Riemann cu $\alpha \le 1$)

Metoda 2: $\left(1+\frac{1}{n}\right)^n < e < \left(1+\frac{1}{n}\right)^{n+1} \Rightarrow$

$$\frac{1}{\left(1+\frac{1}{n}\right)^n} > \frac{1}{e} > \frac{1}{\left(1+\frac{1}{n}\right)^{n+1}} \quad \Big| \cdot \left(1+\frac{1}{n}\right)^n \Rightarrow$$

$$1 > \frac{1}{e}\cdot\left(1+\frac{1}{n}\right)^n > \frac{1}{1+\frac{1}{n}} \Rightarrow 1 > \frac{u_{n+1}}{u_n} > \frac{n}{n+1}$$

$$\Rightarrow \frac{u_{n+1}}{u_n} \ge \frac{\frac{1}{n+1}}{\frac{1}{n}} = \frac{v_{n+1}}{v_n}, \text{ unde } v_n = \frac{1}{n} \Rightarrow \text{cf. criteriului}$$

al dublei de comparație și cum seria $\sum_{n=1}^{\infty} \frac{1}{n}$ este divergentă

$\Rightarrow \sum_{n=1}^{\infty} u_n$ este divergentă

(23.) $\sum_{n=0}^{\infty} \frac{3n^3-n^2+1}{n!} \cdot a^n$, $a>0$

Vom aplica criteriul raportului (*)

$$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to+\infty} \frac{3(n+1)^3-(n+1)^2+1}{(n+1)!\,(n+1)} \cdot a^{n+1} \cdot \frac{n!}{(3n^3-n^2+1)\cdot a^n} =$$

$$= \lim_{n\to+\infty} \frac{a\,[3(n+1)^3-(n+1)^2+1]}{(n+1)(3n^3-n^2+1)} = 0$$, deoarece gradul numitorului este mai mare decât gradul numărătorului.

$\Rightarrow$ cum $\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = 0$ (*) $\Rightarrow$ șirul este convergent.

(24.) $\sum_{n=1}^{\infty} a^n \left(1+\frac{1}{n}\right)^n$, $a>0$

Vom aplica criteriul rădăcinii (Cauchy) (*)

$$\lim_{n\to+\infty} \sqrt[n]{u_n} = \lim_{n\to+\infty} a\underbrace{\left(1+\frac{1}{n}\right)}_{\downarrow 1} = a$$

Dacă $a<1$ (*) $\Rightarrow$ seria este convergentă.

Dacă $a>1$ (*) $\Rightarrow$ seria este divergentă.

Dacă $a=1 \Rightarrow u_n = \left(1+\frac{1}{n}\right)^n \Rightarrow$

$\lim_{n\to+\infty} u_n = \lim_{n\to+\infty} \left(1+\frac{1}{n}\right)^n = e \neq 0 \Rightarrow$ cf. crit. necesar seria este divergentă.

(25.) $\sum_{n=1}^{\infty} \left(b \cdot \frac{a+n}{a+n-1}\right)^n$

Aplicăm criteriul rădăcinii (Cauchy) (*)

$$\lim_{n\to+\infty} \sqrt[n]{u_n} = \lim_{n\to+\infty} \underbrace{b \cdot \frac{a+n}{a+n-1}}_{\downarrow 1} = b$$

Dacă $b<1$ (*) $\Rightarrow$ seria este convergentă.

Dacă $b>1$ (*) $\Rightarrow$ seria este divergentă.

Dacă $b=1 \Rightarrow$ seria devine:

$$u_n = \left(\frac{a+n}{a+n-1}\right)^n$$

$$\lim_{n\to+\infty} u_n = \lim_{n\to+\infty} \left(1+\frac{1}{a+n-1}\right)^n = \lim_{n\to+\infty} \left[\left(1+\frac{1}{a+n-1}\right)^{a+n-1}\right]^{\frac{n}{a+n-1}} =$$

$$= e^{\lim_{n\to+\infty} \frac{n}{a+n-1}} = e \neq 0 \Rightarrow$$ cf. crit. necesar seria este divergentă.

(26.) $\sum_{n=1}^{\infty} \left(a \cdot \frac{n^2+n+1}{n^2}\right)^n$, $a>0$

Aplicăm criteriul radicalului $\Rightarrow$

$$\lim_{n\to+\infty} \sqrt[n]{u_n} = \lim_{n\to+\infty} a \cdot \underbrace{\frac{n^2+n+1}{n^2}}_{\downarrow 1} = a$$

Dacă $a<1 \Rightarrow$ seria este convergentă

Dacă $a>1 \Rightarrow$ seria este divergentă

Dacă $a=1 \Rightarrow$ seria devine:

$$u_n = \left(\frac{n^2+n+1}{n^2}\right)^n \Rightarrow$$

$$\lim_{n\to+\infty} u_n = \underbrace{\left[\left(1+\frac{n+1}{n^2}\right)^{\frac{n^2}{n+1}}\right]}_{\downarrow e}^{\frac{(n+1)\cdot n}{n^2}} = e^{\lim\limits_{n\to+\infty} \frac{(n+1)n}{n^2}} = e \neq 0 \Rightarrow$$

conform criteriului e necesar, seria este cu divergentă

(27.) $\sum_{n=1}^{\infty} \frac{a^n}{\sqrt{n!}}$, $a>0$

Aplicăm criteriul raportului. (*)

$$u_n = \frac{a^n}{\sqrt{n!}}, a>0 \Rightarrow \lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \frac{\lim a^{n+1}}{\sqrt{(n+1)!}} \cdot \frac{\sqrt{n!}}{a^n} = \lim_{n\to+\infty} \frac{a}{\sqrt{n+1}} \overset{a>0}{=} 0<1 \overset{(*)}{\Rightarrow}$$

seria este convergentă.

(28.) $\sum_{n=1}^{\infty} \left[\frac{(2n-1)!!}{(2n)!!}\right]^2 \cdot \frac{1}{n}$

Vom folosi criteriul lui Raabe-Duhamel (*)

$$\lim_{n\to+\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to+\infty} n\left\{\left[\frac{(2n-1)!!}{(2n)!!}\right]^2 \cdot \frac{1}{n} \cdot \left[\frac{(2n+1)!!\,(2n+2)}{(2n-1)!!\,(2n+1)}\right]^2 \cdot (n+1) - 1\right\}$$

$$= \lim_{n\to+\infty} n\left[\left(\frac{2n+2}{2n+1}\right)^2 \cdot \frac{n+1}{n} - 1\right] = \lim_{n\to+\infty} \frac{(2n+2)^2\cdot(n+1) - (2n+1)^2\cdot n}{(2n+1)^2} =$$

$$= \lim_{n\to+\infty} \frac{(4n^2+8n+4)(n+1) - (4n^2+4n+1)\cdot n}{(2n+1)^2} = \lim_{n\to+\infty} \frac{4n^3+8n^2+4n+4n^2+8n+4-4n^3-4n^2-n}{(2n+1)^2}$$

$$= \lim_{n\to+\infty} \frac{8n^2+11n+4}{(2n+1)^2} = \frac{8}{4} = 2 > 1 \overset{(*)}{\Rightarrow}$$ seria este convergentă.

(29.) $\sum_{n=1}^{\infty} \left(\frac{1\cdot 4\cdot 7\cdot \ldots\cdot (3n-2)}{3\cdot 6\cdot 9\cdot \ldots\cdot (3n)}\right)^2$

Vom folosi criteriul lui Raabe-Duhamel (*)

$$\lim_{n\to+\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to+\infty} n\left[\left(\frac{1\cdot 4\cdot 7\cdot \ldots\cdot (3n-2)}{3\cdot 6\cdot 9\cdot \ldots\cdot (3n)}\right)^2 \cdot \left(\frac{3\cdot 6\cdot 9\cdot \ldots\cdot (3n+3)}{1\cdot 4\cdot 7\cdot \ldots\cdot (3n+1)}\right)^2 - 1\right] =$$

$$= \lim_{n\to+\infty} n\left[\left(\frac{3n+3}{3n+1}\right)^2 - 1\right] = \lim_{n\to+\infty} n\cdot \frac{9(n+1)^2 - (3n+1)^2}{(3n+1)^2} =$$

$$= \lim_{n\to+\infty} n\cdot \frac{9n^2+18n+9-9n^2-6n-1}{(3n+1)^2} = \lim_{n\to+\infty} n\cdot \frac{12n+8}{(3n+1)^2} = \frac{12}{9} = \frac{4}{3} > 1 \Rightarrow$$

seria este convergentă.

(30.) $\sum_{n=2}^{\infty} \frac{1}{(\ln\ln n)^2}$

Voi aplica criteriul logaritmic. (*)

$$\lim_{n\to+\infty} \frac{\ln \frac{1}{u_n}}{\ln n} = \lim_{n\to+\infty} \frac{\ln\left((\ln\ln n)^2\right)}{\ln n} = \lim_{n\to+\infty} \frac{2\ln(\ln(\ln n))}{\ln n} =$$

$$\overset{L.H.}{=} \lim_{n\to+\infty} 2 \cdot \frac{1}{\ln(\ln n)} \cdot \frac{1}{\ln n} \cdot \frac{1}{n} = 0 < 1 \overset{(*)}{\Rightarrow}$$ seria este divergentă

($\ln(\ln n) \to 0$, $\ln n \to 0$)

(31.) $\sum_{n=1}^{\infty} \frac{\ln n}{\sqrt[3]{n^4}}$

Voi aplica criteriul logaritmic. (*)

$$\lim_{n\to+\infty} \frac{\ln \frac{1}{u_n}}{\ln n} = \lim_{n\to+\infty} \frac{\ln \frac{\sqrt[3]{n^4}}{\ln n}}{\ln n} = \lim_{n\to+\infty} \frac{\ln \sqrt[3]{n^4} - \ln(\ln n)}{\ln n} =$$

$$= \lim_{n\to+\infty} \left[\frac{4}{3} \cdot \frac{\ln n}{\ln n} - \frac{\ln(\ln n)}{\ln n}\right] = \frac{4}{3} - \lim_{n\to+\infty} \frac{\ln(\ln n)}{\ln n} \overset{L.H.}{=}$$

$$\frac{4}{3} - \lim_{n\to+\infty} \frac{1}{n} \cdot \frac{1}{\ln n} \cdot n = \frac{4}{3} > 1 \overset{(*)}{\Rightarrow}$$ seria este convergentă.

($\ln n \to 0$)

(32.) $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$

Aplicăm criteriul logaritmic. (*)

$$\lim_{n\to+\infty} \frac{\ln \frac{1}{u_n}}{\ln n} = \lim_{n\to+\infty} \frac{\ln n^{\sqrt{n}}}{\ln n} = \lim_{n\to+\infty} \frac{\sqrt{n} \cdot \ln n}{\ln n} = \lim_{n\to+\infty} \sqrt{n} = +\infty > 1$$

$\Rightarrow$ seria este convergentă.

(33.) $\sum_{n=2}^{\infty} \frac{1}{n \ln^2 n}$

Aplicăm criteriul integral (*)

Fie $f: [2, +\infty) \to (0, +\infty)$, $f(x) = \frac{1}{x \ln^2 x}$, funcție descrescătoare (produs de fct. descrescătoare)

Fie $a_n = \int_2^n f(x)\,dx = \int_2^n \frac{1}{x \ln^2 x}\,dx = -\frac{1}{\ln x}\Big|_2^n =$

$= \frac{1}{\ln 2} - \frac{1}{\ln n} \Rightarrow \lim_{n\to+\infty} a_n = \frac{1}{\ln 2} \Rightarrow (a_n)_{n\geq 2}$ este convergentă

$\Rightarrow$ cf. criteriului, seria este convergentă.

(36.) $\sum_{n=1}^{\infty} a^{\ln n}$, $a>0$

Aplicăm criteriul logaritmic.

$$\lim_{n\to+\infty} \frac{\ln \frac{1}{u_n}}{\ln n} = \lim_{n\to+\infty} \frac{\ln a^{-\ln n}}{\ln n} = \lim_{n\to+\infty} \frac{-\ln n \cdot \ln a}{\ln n} = -\ln a$$

Dacă $-\ln a > 1 \Leftrightarrow \ln a < -1 \Leftrightarrow a < \frac{1}{e} \Rightarrow$ seria este convergentă

Dacă $-\ln a < 1 \Leftrightarrow a > \frac{1}{e} \Rightarrow$ seria este divergentă

Dacă $-\ln a = 1 \Leftrightarrow a = \frac{1}{e} \Rightarrow$ seria devine:

$u_n = a^{\ln n} = \left(\frac{1}{e}\right)^{\ln n} = e^{-\ln n} = \frac{1}{n}$, care este seria armonică

$\Rightarrow$ seria este divergentă.

(37.) $\sum_{n=1}^{\infty} \frac{\sqrt{n!}}{(2+\sqrt{1})(2+\sqrt{2})\cdot\ldots\cdot(2+\sqrt{n})}$

Vom aplica criteriul lui Raabe-Duhamel.

$$\lim_{n\to+\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to+\infty} n\left(\frac{\sqrt{n!}}{(2+\sqrt{1})(2+\sqrt{2})\cdots(2+\sqrt{n})} \cdot \frac{(2+\sqrt{1})(2+\sqrt{2})\cdots(2+\sqrt{n+1})}{\sqrt{n!}\cdot\sqrt{n+1}} - 1\right)$$

$$= \lim_{n\to+\infty} n\cdot\left(\frac{2+\sqrt{n+1}}{\sqrt{n+1}} - 1\right) = \lim_{n\to+\infty} n\cdot\frac{2+\sqrt{n+1}-\sqrt{n+1}}{\sqrt{n+1}} = \lim_{n\to+\infty}\frac{2n}{\sqrt{n+1}} = +\infty > 1$$

$\Rightarrow$ seria este convergentă.

(38.) $\sum_{n=1}^{\infty} \left(\frac{1}{\ln n}\right)^{\ln(\ln n)}$

Aplicăm criteriul logaritmic.

$$\lim_{n\to+\infty} \frac{\ln \frac{1}{u_n}}{\ln n} = \lim_{n\to+\infty} \frac{\ln(\ln(\ln n))}{\ln n} \lim_{n\to+\infty} \frac{\ln\left[(\ln n)^{\ln(\ln n)}\right]}{\ln n} =$$

$$= \lim_{n\to+\infty} \frac{\ln(\ln n)\cdot\ln(\ln n)}{\ln n} = \lim_{n\to+\infty} \frac{\ln^2(\ln n)}{\ln n} \overset{L.H.}{=}$$

$$\lim_{n\to+\infty} 2\ln(\ln n)\cdot\frac{1}{\ln n}\cdot\frac{1}{n}\cdot n = \lim_{n\to+\infty}\frac{2\ln(\ln n)}{\ln n} \overset{L.H.}{=}$$

$$\lim_{n\to+\infty} 2\cdot\frac{1}{\ln n}\cdot\frac{1}{n}\cdot n = \lim_{n\to+\infty}\frac{2}{\ln n} = 0 < 1 \Rightarrow \text{seria este divergentă}$$

(39.) $\sum_{n=1}^{\infty} \left[\sqrt{(n+1)(n+a)} - n\right]^n$, $a>0$

Aplicăm criteriul rădăcinii.

$$\lim_{n\to+\infty} \sqrt[n]{u_n} = \lim_{n\to+\infty}\left(\sqrt{(n+1)(n+a)} - n\right) = \lim_{n\to+\infty} \frac{(n+1)(n+a)-n^2}{\sqrt{(n+1)(n+a)}+n} = \lim_{n\to+\infty}\frac{n^2+an+n+a-n^2}{\sqrt{(n+1)(n+a)}+n}$$

$$= \lim_{n\to+\infty}\frac{n(a+1)+a}{\sqrt{(n+1)(n+a)}+n} = \frac{a+1}{2}$$

Dacă $\frac{a+1}{2} < 1 \Leftrightarrow a < 1 \Rightarrow$ seria este convergentă

Dacă $\frac{a+1}{2} > 1 \Leftrightarrow a > 1 \Rightarrow$ seria este divergentă

Dacă $\frac{a+1}{2} = 1 \Leftrightarrow a = 1 \Rightarrow$ seria devine:

$u_n = \left[\sqrt{(n+1)^2} - n\right]^n = (n+1-n)^n = 1 \neq 0 \Rightarrow$ cf. crit. necesar seria este divergentă.

(40.) $\sum_{n=1}^{\infty} \left(\frac{an+b}{cn+d}\right)^n$, $a,b,c,d>0$

Aplicăm criteriul rădăcinii (Cauchy).

$\lim_{n\to+\infty} \sqrt[n]{u_n} = \lim_{n\to+\infty} \frac{an+b}{cn+d} = \frac{a}{c} = e$

Dacă $\frac{a}{c} > 1 \Rightarrow$ seria este divergentă

Dacă $\frac{a}{c} < 1 \Rightarrow$ seria este convergentă

Dacă $\frac{a}{c} = 1 \Rightarrow$ seria devine:

$u_n = \left(\frac{an+b}{an+d}\right)^n = \left(1 + \frac{b-d}{an+d}\right)^n$

Dacă $b \neq d \Rightarrow$

$\lim_{n\to+\infty} u_n = \lim_{n\to+\infty} \left(1 + \frac{b-d}{an+d}\right)^n = \lim_{n\to+\infty} \left[\left(1+\frac{b-d}{an+d}\right)^{\frac{an+d}{b-d}}\right]^{\frac{n(b-d)}{an+d}}$

$= e^{\lim_{n\to+\infty} \frac{n(b-d)}{an+d}} = e^{\frac{b-d}{a}} \neq 0 \Rightarrow$ cf. crit. necesar seria este divergentă

Dacă $b=d \Rightarrow$

$u_n = 1 \neq 0 \Rightarrow$ cf. crit. necesar seria este divergentă.

(41.) $\sum_{n=1}^{\infty} tg^n\left(a + \frac{b}{n^2}\right)$, $0 < a < \frac{\pi}{2}$, $b \in \mathbb{R}$

Aplicăm criteriul rădăcinii (Cauchy) $\Rightarrow$

$\lim_{n\to+\infty} \sqrt[n]{u_n} = \lim_{n\to+\infty} tg\left(a + \frac{b}{n^2}\right) = tg\,a$

Dacă $tg\,a > 1 \Leftrightarrow a > \frac{\pi}{4} \Rightarrow$ seria este divergentă.

Dacă $tg\,a < 1 \Leftrightarrow a < \frac{\pi}{4} \Rightarrow$ seria este convergentă.

Dacă $tg\,a = 1 \Leftrightarrow a = \frac{\pi}{4} \Rightarrow$

$\lim_{n\to+\infty} u_n = \lim_{n\to+\infty} tg^n\left(\frac{\pi}{4} + \frac{b}{n^2}\right) = \left(\frac{tg\frac{\pi}{4} + tg\frac{b}{n^2}}{1 - tg\frac{\pi}{4}\cdot tg\frac{b}{n^2}}\right)^n =$

$\lim_{n\to+\infty} \left(1 + \frac{2tg\frac{b}{n^2}}{1 - tg\frac{b}{n^2}}\right)^n = \lim_{n\to+\infty} \left[\left(1 + \frac{2tg\frac{b}{n^2}}{1 - tg\frac{b}{n^2}}\right)^{\frac{1-tg\frac{b}{n^2}}{2tg\frac{b}{n^2}}}\right]^{\frac{2n\cdot tg\frac{b}{n^2}}{1-tg\frac{b}{n^2}}}$

$= \lim_{n\to+\infty} e^{\frac{2n\,tg\frac{b}{n^2}}{1-tg\frac{b}{n^2}}} = e^{\lim_{n\to+\infty} \frac{2n}{1-tg\frac{b}{n^2}} \cdot \frac{tg\frac{b}{n^2}}{\frac{b}{n^2}} \cdot \frac{b}{n^2}} =$ ($\frac{tg\frac{b}{n^2}}{\frac{b}{n^2}} \to 1$)

$= e^{\lim_{n\to+\infty} \frac{2b}{n(1-tg\frac{b}{n^2})}} = e^0 = 1 \neq 0 \Rightarrow$ cf. crit. necesar

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(42.) $\sum_{n=2}^{\infty} (n-1)^2 \cdot \arcsin \frac{\pi}{2^{n+1}}$

Aplicăm criteriul raportului.

$$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to+\infty} \frac{n^2}{(n+1)^2} \cdot \frac{\arcsin \frac{\pi}{2^{n+2}}}{\arcsin \frac{\pi}{2^{n+1}}} = \lim_{n\to+\infty} \frac{\arcsin \frac{\pi}{2^{n+2}}}{\frac{\pi}{2^{n+2}}} \cdot \frac{\frac{\pi}{2^{n+1}}}{\arcsin \frac{\pi}{2^{n+1}}} \cdot \frac{\frac{\pi}{2^{n+2}}}{\frac{\pi}{2^{n+1}}}$$

($\frac{n^2}{(n+1)^2} \to 1$; $\frac{\arcsin \frac{\pi}{2^{n+2}}}{\frac{\pi}{2^{n+2}}} \to 1$; $\frac{\frac{\pi}{2^{n+1}}}{\arcsin \frac{\pi}{2^{n+1}}} \to 1$)

$$= \lim_{n\to+\infty} \frac{2^{n+1}}{2^{n+2}} = 1/2 < 1 \Rightarrow \text{seria este convergentă.}$$

(43.) $\sum_{n=1}^{\infty} 2^{n-1} \cdot \operatorname{tg}^2 \frac{a}{2^n} \cdot \operatorname{tg} \frac{a}{2^{n-1}}$, $a>0$

Aplicăm criteriul raportului.

$$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to+\infty} \frac{2^n \cdot \operatorname{tg}^2 \frac{a}{2^{n+1}} \cdot \operatorname{tg} \frac{a}{2^n}}{2^{n-1} \cdot \operatorname{tg}^2 \frac{a}{2^n} \cdot \operatorname{tg} \frac{a}{2^{n-1}}} = \lim_{n\to+\infty} 2 \cdot \left(\frac{\operatorname{tg} \frac{a}{2^{n+1}}}{\frac{a}{2^{n+1}}} \right)^2 \cdot \left(\frac{a}{2^{n+1}} \right)^2 \cdot \frac{\frac{a}{2^n}}{\operatorname{tg} \frac{a}{2^n}} \cdot \frac{2^n}{a} \cdot \frac{\frac{a}{2^{n-1}}}{\operatorname{tg} \frac{a}{2^{n-1}}} \cdot \frac{2^{n-1}}{a}$$

($\left(\frac{\operatorname{tg} \frac{a}{2^{n+1}}}{\frac{a}{2^{n+1}}} \right)^2 \to 1$; $\frac{\frac{a}{2^n}}{\operatorname{tg} \frac{a}{2^n}} \to 1$; $\frac{\frac{a}{2^{n-1}}}{\operatorname{tg} \frac{a}{2^{n-1}}} \to 1$)

$$= \lim_{n\to+\infty} \frac{2 \cdot 2^{n-1} \cdot 2^n}{2^{2n+2}} = \lim_{n\to+\infty} \frac{2^{2n}}{2^{2n+2}} = 1/4 < 1$$

$\Rightarrow$ seria este convergentă

(44.) $\sum_{n=1}^{\infty} \frac{1}{4^n \cdot \cos^2 \frac{a}{2^n}}$

Aplicăm criteriul raportului. ($\cos^2 \frac{a}{2^n} \to 1$)

$$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to+\infty} \frac{4^n \cdot \cos^2 \frac{a}{2^n}}{4^{n+1} \cdot \cos^2 \frac{a}{2^{n+1}}} = 1/4 < 1 \Rightarrow \text{seria este convergentă}$$

(45.) $\sum_{n=1}^{\infty} \frac{n^p \cdot \ln n}{n!}$, $p \in \mathbb{R}$

$$u_n = \frac{n^p \cdot \ln n}{n!} < \frac{n^p \cdot n}{n!} = \frac{n^{p+1}}{n!} = v_n$$

Aplicăm criteriul raportului pt. seria $\sum_{n=1}^{\infty} v_n$

$$\lim_{n\to+\infty} \frac{v_{n+1}}{v_n} = \lim_{n\to+\infty} \frac{(n+1)^{p+1}}{n!(n+1)} \cdot \frac{n!}{n^{p+1}} = \lim_{n\to+\infty} \left(\frac{n+1}{n} \right)^{p+1} \cdot \frac{1}{n+1} = 0 < 1$$

($\left(\frac{n+1}{n} \right)^{p+1} \to 1$; $\frac{1}{n+1} \to 0$)

$\Rightarrow$ seria $\sum_{n=1}^{\infty} v_n$ este convergentă și cum $u_n < v_n$ $(\forall) n \in \mathbb{N}^*$

$\Rightarrow$ cf. primului criteriu de comparație seria $\sum_{n=1}^{\infty} u_n$ este convergentă.